LITERATURE MAPPING REPORT

Research-to-Implementation Mapping



How 20 team-selected papers and 44 further verified sources shaped FraudLens: what we learned, what we adopted, and where each idea lives in the system

20 team papers | 44 additional papers | 11 industry & regulatory sources | 16 techniques traced

PROJECT
FraudLens: AI-Based Credit Card Fraud Detection

DOMAIN
Cyber Security / Forensics · Machine Learning / Data Security

TEAM
[Team member names]

GUIDE
[Guide name]

INSTITUTION
[College / University]

DATE
September 2026

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1 How to read this report

This report shows, paper by paper and technique by technique, **what we learned from the literature, what we adopted or deliberately rejected, and where it is implemented** in the FraudLens code and experiments. The main Project Report explains the system; this document is the audit trail from research to implementation.

20 Team-selected papers — 19 fully verified

44 Additional papers — 42 fully verified

11 Industry / regulatory — sources with dates

16 Techniques traced — to code + experiment

7 Experiments (E1-E7) — provide the evidence

Table 1. Experiments referenced as evidence throughout this report

Code	Experiment	Question it answers
E1	Model benchmark	Which model family works best on the same chronological split, and how fast is it?
E2	Imbalance handling	No balancing vs class weighting vs undersampling vs SMOTE: what really changes?
E3	Evaluation protocol	How much do common shortcuts (SMOTE before split, random split, accuracy) inflate results?
E4	Ablation	Do the behavioural features and each hybrid component actually help?
E5	Unseen fraud	Does the anomaly layer catch a fraud type the supervised models never saw?
E6	ULB benchmark	How does the system do on the European dataset most papers use?

Code	Experiment	Question it answers
E7	Business analysis	Money saved, risk tiers, stability over time, drift, statistical significance

Verification

Every citation was checked against Crossref, OpenAlex, Semantic Scholar or the publisher page. Where the team's list differed from the published record (title, year or venue), the correction is listed in Section 2 and the corrected form is used everywhere else.

2 Corrections to the team's paper list

Table 2. Differences between the team's list and the published record

#	As listed by the team	Verified correction
#7	Enhancing Fraud Detection ... using XGBoost and SMOTE (2024)	Full title ends with ': A Comparative Study'
#8	Design and Empirical Analysis ... (IEEE-2025)	Published at ICBAIE **2024** (added to IEEE Xplore in 2025)
#10	... Deep Learning Method Using Data Balance Tools (IEEE-2021)	ICCSMT **2020** (added to Xplore in 2021)
#11	An Assessment of Streaming Active Learning Strategies ... (IEEE-2018)	IEEE DSAA **2017** (Xplore 2018); journal extension in IJDSA 2018
#13	XAI for Fraud Detection: An Attention-Based Ensemble ... (IEEE-2025)	IEEE TIFS vol. 21 (**2026** volume, online Dec 2025); title ends '... for Reliable Decision-Making'
#14	Graph Neural Network-Based Fraud Detection in Credit Card Transactions (IEEE-2025)	Real title: 'Similarity-Navigated Graph Neural Network-Based Fraud Detection in Credit Card Transactions Optimized with the Greylag Goose Algorithm' (ICMLAS 2025)
#16	Credit Card Fraud Detection using ML: Methods, Challenges and Solutions (IEEE-2025)	BITCON **2024**
#17	Advancement in Credit Card Fraud Detection ... (IEEE-2025)	ICSCAI **2024**
#18	Advanced Machine Learning Techniques for Credit Card Fraud Detection (IEEE-2024)	Title ends with ': A Comprehensive Study' (ICOSEC 2024)
#20	Credit Card Fraud Detection in the Era of Disruptive Technologies (IEEE-2023)	**Not IEEE**: Journal of King Saud University - Computer and Information Sciences 35(1), Elsevier, 2023

Titles #4 and #18 are generic (several papers share them); the one matching the stated year and content was selected, and the alternatives are recorded in docs/sources/literature.json.

3 The team's 20 papers: what we learned and where it lives

Table 3. Paper-by-paper mapping of the team's literature set

#	Paper (verified)	Approach and dataset	Reported result	What we learned	What we adopted / changed	Where in FraudLens
1	**A Novel Deep Learning Framework for Credit Card Fraud Detection**. Banoth & Madhavi, 2024 13th International Conference on System Modeling & Advancement in Research Trends (SMART), 2024.	Tabular transaction records converted to small images with a proposed 'text2IMG' method; convolutional neural network with inverse-frequency class weights to handle imbalance; compared against other DL and ML methods. *Data:* A Kaggle credit-card fraud dataset (not named more precisely in the abstract)	Accuracy 99.87% (abstract)	Deep learning can learn fraud patterns once tabular data is reshaped, but the paper reports accuracy only.	Built a 1D-CNN baseline (features as a 1-channel sequence) and judged it by PR-AUC, not accuracy. **Our CNN: PR-AUC 0.933, recall 0.827.**	fraudlens/deep.py, E1
2	**Fraud Shield: Credit Card Fraud Detection with Ensemble and Deep Learning**. Menon et al., 2024 4th International Conference on Pervasive Computing and Social Networking (ICPCSN), 2024.	Ensemble learning (Voting, Stacking) and deep learning (CNN, LSTM); addresses data imbalance and feature-selection inefficiencies. *Data:* IEEE-CIS Fraud Detection dataset (Kaggle / Vesta)	No numeric results in the abstract	Combining diverse models (voting/stacking) is more robust than one model, at the cost of complexity.	Kept the stacking idea but with light, explainable learners (XGB + RF + IF) and a logistic meta-learner. **Our hybrid: PR-AUC 0.979, recall 0.925.**	fraudlens/hybrid.py, E4
3	**Online Transaction Anomaly Detection Model for Credit Card Usage Using Machine Learning Classifiers**. Jayasingh & Sri, 2023 International Conference on Emerging Smart Computing and Informatics (ESCI), 2023.	Proposed Transaction Anomaly Detection (TAD) model; several ML classifiers compared on accuracy and recall; XGBoost classifier selected as best. *Data:* Kaggle credit-card dataset with 284,807 labelled online transactions (i.e., the ULB/Worldline European dataset)	XGB classifier: accuracy 99.96%, recall 83% (abstract)	XGBoost is the strongest classical learner on card data; 99.96% accuracy hid a recall of only 83%.	XGBoost is the main classifier; recall, PR-AUC and MCC are reported instead of accuracy. **Our XGB: PR-AUC 0.978, recall 0.937.**	fraudlens/hybrid.py, E1, E6
4	**Credit Card Fraud Detection and Analysis**. Pawar et al., 2023 14th International Conference on Computing Communication and Networking Technologies (ICCCNT), 2023.	Machine-learning algorithms applied to detect anomalous (fraudulent) credit-card transactions; abstract is generic. *Data:* Not stated in the abstract	No numeric results in the abstract	A plain ML classifier is a useful baseline but does not adapt to new fraud or explain decisions.	Included Logistic Regression, Decision Tree and a legacy rule engine as baselines. **Our LR: PR-AUC 0.441, recall 0.735.**	fraudlens/rules.py, E1
5	**Detection of Credit Card Fraud Using Resampling and Boosting Technique**. Jose et al., 2023 Advanced Computing and Communication Technologies for High Performance Applications (ACCTHPA), 2023.	SMOTE oversampling; Extra Trees, Gradient Boosting, Decision Tree, Random Forest, each also combined with AdaBoost; evaluated with accuracy, recall, precision, F1. *Data:* Sparkov imbalanced synthetic credit-card dataset (named attributes), explicitly contrasted with the European (ULB) Kaggle dataset	No numeric results in the abstract	Same Sparkov dataset; SMOTE + boosting is the usual recipe for imbalance.	Adopted Sparkov as the primary dataset and tested SMOTE against class weighting and undersampling.	fraudlens/models.py, E2

#	Paper (verified)	Approach and dataset	Reported result	What we learned	What we adopted / changed	Where in FraudLens
6	**A Machine Learning Approach for Credit Card Fraud Detection in Massive Datasets Using SMOTE and Random Sampling**. B et al., 2024 IEEE Recent Advances in Intelligent Computational Systems (RAICS), 2024.	ML fraud-detection model with user separation, SMOTE and random sampling for the large, imbalanced dataset. *Data:* 'IEEE CIS Credit Fraud Detection' dataset from Kaggle	No numeric results in the abstract	Large, imbalanced data needs sampling and per-user separation to scale.	Per-card (per-user) behavioural state in a streaming feature store; undersampling arm in E2.	fraudlens/features.py, E2
7	**Enhancing Fraud Detection in Credit Card Transactions using XGBoost and SMOTE: A Comparative Study**. Gupta et al., 2024 IEEE International Conference on Smart Power Control and Renewable Energy (ICSPCRE), 2024.	XGBoost combined with SMOTE, compared against established ML methods. *Data:* Not named in the abstract	XGBoost + SMOTE: precision 99.94%, accuracy 99.9684% (abstract)	XGBoost + SMOTE is popular; near-perfect numbers (99.94% precision) raise leakage concerns.	Re-ran XGBoost + SMOTE under four evaluation protocols to show how leakage inflates results.	fraudlens/pipeline.py, E3
8	**Design and Empirical Analysis of Data-Driven Machine Learning Models for Credit Card Fraud Detection**. Wang et al., 2024 5th International Conference on Big Data, 2024.	Logistic Regression, Support Vector Machine, Decision Tree. *Data:* Transaction records from 10,000 users (source not named)	All three models exceed 98% precision; logistic regression best overall accuracy and recall; decision trees best at reducing false positives (abstract)	LR, SVM, DT exceed 98% precision on small data; DT reduces false positives.	LR and DT kept as interpretable baselines on the same chronological split. **Our DT: PR-AUC 0.787, recall 0.743.**	E1
9	**A Hybrid Deep Learning Model For Online Fraud Detection**. Kewei et al., 2021 IEEE International Conference on Consumer Electronics and Computer Engineering (ICCECE), 2021.	Deep-learning model with feature engineering, memory compression, mixed precision training and an ensemble loss. *Data:* IEEE-CIS fraud dataset (Vesta Corporation), over 1 million records	Outperforms Bayes and SVM baselines (abstract; no numbers)	Hybrid DL on IEEE-CIS still depends on feature engineering.	Invested in behavioural feature engineering (27 engineered features) instead of heavier networks.	fraudlens/features.py, E4
10	**Credit Card Fraud Detection via Deep Learning Method Using Data Balance Tools**. Zhang & Huang, 2020 International Conference on Computer Science and Management Technology (ICCSMT), 2020.	CNN and auto-encoder; random under-sampling and SMOTE for balancing. *Data:* ULB (Universite Libre de Bruxelles) PCA-transformed dataset plus a new raw-variable dataset	About 93% accuracy on the ULB dataset; poor performance on the raw dataset unless network complexity is increased; under-sampling improved accuracy more than oversampling (abstract)	Balancing matters as much as the network; undersampling beat oversampling in their CNN.	Four-way imbalance comparison; DL baselines trained with class weights. **Our MLP: PR-AUC 0.960, recall 0.890.**	fraudlens/models.py, fraudlens/deep.py, E2

#	Paper (verified)	Approach and dataset	Reported result	What we learned	What we adopted / changed	Where in FraudLens
11	**An Assessment of Streaming Active Learning Strategies for Real-Life Credit Card Fraud Detection**. Carcillo et al., 2017 IEEE International Conference on Data Science and Advanced Analytics (DSAA), 2017.	Streaming active-learning strategies combined with semi-supervised learning; exploration/exploitation trade-off for querying investigators. *Data:* Real-world dataset of millions of transactions from industrial partner Worldline	Baseline 'High Risk Querying' is a robust strategy and is further improved by combining it with semi-supervised learning (abstract; no numbers)	Real fraud detection is streaming, non-stationary, with investigator feedback.	Time-ordered replay stream, STEP-UP tier and an analyst case queue whose verdicts are stored as labels.	fraudlens/api/, dashboard /monitor and /cases
12	**Deep Learning Techniques for Financial Fraud Detection**. Yang et al., 2022 14th International Conference on Computer Research and Development (ICCRD), 2022.	Review of ML/DL (supervised and unsupervised) methods for financial fraud detection. *Data:* None (review paper)	None (review)	Survey of CNN/RNN/LSTM for financial fraud.	Background for the DL baselines and the future-scope LSTM.	Report ch. 2, fraudlens/deep.py
13	**XAI for Fraud Detection: An Attention-Based Ensemble of CNNs, GNNs, and Confidence-Driven Gating for Reliable Decision-Making**. Chagahi et al., IEEE Transactions on Information Forensics and Security, 2026.	Attention-based ensemble of CNN, GNN, RNN and LSTM; confidence-driven gating; DOWA and IOWA aggregation operators; SHAP-based feature selection. *Data:* Three benchmark fraud datasets (not named in the abstract), balanced and highly imbalanced settings	Surpasses individual classifiers and existing ensembles on all three datasets (abstract; no numbers)	Weight detectors by reliability (confidence gating) and explain with SHAP.	Detector weights are learned by the meta-learner; SHAP reason codes for every decision. **Our hybrid: PR-AUC 0.979, recall 0.925.**	fraudlens/hybrid.py, fraudlens/explain.py
14	**Similarity-Navigated Graph Neural Network-Based Fraud Detection in Credit Card Transactions Optimized with the Greylag Goose Algorithm**. Sundararamaiah et al., 2025 International Conference on Machine Learning and Autonomous Systems (ICMLAS), 2025.	Z-score/Min-Max normalisation; hybrid feature selection with Crayfish Optimization Algorithm + Emperor Penguin Optimizer; Similarity-Navigated GNN classifier; Greylag Goose Optimization. *Data:* Credit Card Fraud Detection (CCF) dataset (Kaggle)	About 99% accuracy (abstract)	Fraud involves relationships (card-merchant-device); GNNs capture them at high cost.	Relationship-aware behavioural features now (new merchant/category for this card); GNN as future work.	fraudlens/features.py; future scope
15	**A Systematic Literature Survey on Credit Card Fraud Detection using Machine Learning Techniques**. Pathak & Beniwal, 2025 5th International Conference on Pervasive Computing and Social Networking (ICPCSN), 2025.	Survey of ML and DL approaches, challenges and solutions for fraud transactions; data-mining techniques; identifies countries most vulnerable to financial fraud. *Data:* None (survey)	None (survey)	Common datasets, techniques and open gaps.	Gap analysis that shaped the objectives.	Report ch. 2
16	**Credit Card Fraud Detection using Machine Learning: Methods, Challenges and Solutions**. Saini et al., 2024 International BIT Conference (BITCON), 2024.	Review of research gaps: data imbalance, real-time processing, scalability, explainability; discusses Dempster-Shafer + Bayesian hybrids, SMOTE, AdaBoost ensembles; argues for scalable, interpretable, adaptive real-time systems. *Data:* None (review)	None (review)	Open challenges: imbalance, real-time, scalability, explainability.	Each challenge has a module: E2 + calibration, real-time API, 67 us/txn feature store, SHAP.	Whole system

#	Paper (verified)	Approach and dataset	Reported result	What we learned	What we adopted / changed	Where in FraudLens
17	**Advancement in Credit Card Fraud Detection: A Comprehensive Review of Machine Learning Techniques**. Chawla et al., 2024 1st International Conference on Sustainable Computing and Integrated Communication in Changing Landscape of AI (ICSCAI), 2024.	Review of ML techniques for card fraud detection: neural networks, logistic regression, random forest, decision trees. *Data:* None (review)	None (review)	Evolution from rules to ML to ensembles.	Evolution storyline and the legacy-rules baseline. **Our RULES: PR-AUC 0.248, recall 0.268.**	Report ch. 2, fraudlens/rules.py
18	**Advanced Machine Learning Techniques for Credit Card Fraud Detection: A Comprehensive Study**. Sonwane et al., 2024 5th International Conference on Smart Electronics and Communication (ICOSEC), 2024.	Random Forest, Decision Tree and Artificial Neural Network compared; emphasis on feature engineering. *Data:* A publicly available credit-card dataset (not named in the abstract)	No numeric results in the abstract	RF, DT, ANN compared with an emphasis on feature engineering.	Random Forest as the second opinion; engineered features. **Our RF: PR-AUC 0.943, recall 0.851.**	fraudlens/hybrid.py, fraudlens/features.py
19	**Deep Learning for Credit Card Fraud Detection: A Review of Algorithms, Challenges, and Solutions**. Mienye & Jere, IEEE Access, 2024.	Review and performance comparison of CNN, simple RNN, LSTM and GRU; discussion of suitable performance metrics and training challenges. *Data:* Review; experimental comparison uses 'a real-world dataset' (not named in the abstract)	Experimental results indicate robustness of DL architectures (abstract; no numbers)	DL review: CNN/RNN/LSTM/GRU and the right metrics for imbalance.	PR-AUC/MCC metrics; sequence signal via velocity windows; LSTM as future work.	fraudlens/metrics.py; future scope
20	**Credit card fraud detection in the era of disruptive technologies: A systematic review**. Cherif et al., Journal of King Saud University - Computer and Information Sciences, 2023.	Systematic review categorised by topic (class imbalance, feature engineering, etc.) and by ML technology (traditional vs deep learning); identifies limited DL investigation and need for big-data / large-scale ML / cloud approaches. *Data:* None (systematic review of 40 articles, 2015-2021)	None (review)	Modern fraud detection needs big-data / cloud-ready, scalable deployment.	API-first design, PostgreSQL-ready decision log, Docker deployment; streaming as future scope.	fraudlens/api/, Dockerfile

4 Technique traceability matrix

Table 4. Technique -> sources -> code -> evidence (T# = team paper number)

Technique	Source papers	Implementation	Evidence
Gradient boosting (XGBoost) as main classifier	T3, T7, Chen & Guestrin 2016, Grinsztajn 2022	hybrid.py	E1: XGBoost PR-AUC 0.978 vs Logistic Regression 0.441
Random Forest second opinion	T5, T18, Breiman 2001, Cavdar 2026	hybrid.py	E1: RF PR-AUC 0.943; used in the stack
Isolation Forest anomaly layer	Liu 2008, Carcillo 2021, Shanaa 2025	hybrid.py	IF alone PR-AUC 0.421; stacking weight -0.09, so it also gets its own novelty route
Safety layer: policy rules + novelty alarm	RBI Master Direction 2021 (paras 34, 36-37), Carcillo 2021, Bahnsen 2016	policy.py	0.81 extra step-ups/day on the test period; catches card testing and cloned cards in replays
Stacked meta-learner (learned weights)	T2, T13, Wolpert 1992	hybrid.fit_meta	E4: stack PR-AUC 0.979 vs XGB alone 0.978
Behavioural aggregation features	Whitrow 2009, Bahnsen 2016, T9, T18	features.py	E4: PR-AUC raw fields 0.825 -> behavioural 0.978
Imbalance handling comparison	T5, T6, T7, T10, Chawla 2002, van den Goorbergh 2022	models.make_model	E2: XGB PR-AUC with SMOTE 0.980 vs class weight 0.978
Calibrated probability + cost-based thresholds	Elkan 2001, Niculescu-Mizil 2005, Vanderschueren 2022, Hoppner 2022	hybrid.py, metrics.py	E7: 98.1% of fraud losses avoided net of alert costs
Leakage-free evaluation	Santos 2018, Kaufman 2012, Hayat 2025, Dal Pozzolo 2018	data.py, pipeline.py	E3 (ULB): F1 1.000 with SMOTE-before-split vs 0.803 honest
PR-AUC / MCC instead of accuracy	Saito 2015, Chicco 2020, T19	metrics.py	'Always genuine' scores 99.40% accuracy on Sparkov test data
SHAP reason codes	T13, Lundberg 2017/2020, Psychoula 2021, Iqbal 2025	explain.py	Every API decision ships its top reasons
Deep-learning baselines	T1, T2, T10, T12, T19	deep.py	E1: MLP 0.960, 1D-CNN 0.933 vs XGBoost 0.978
Real-time API + online feature store	T11, T20, Carcillo 2018 (SCARFF)	api/, features.py	Hybrid model latency p50 209.39 ms per transaction
Human-in-the-loop feedback	T11, Carcillo 2018, Dal Pozzolo 2015	api/db.py, /cases	Review verdicts stored with the audit record

Technique	Source papers	Implementation	Evidence
Graph relationships	T14, Liu 2021 (PC-GNN)	features.py (novelty)	Future work: card-merchant graph
Sequence models	T19, Jurgovsky 2018	features.py (windows)	Future work: LSTM on card histories

5 Additional literature beyond the team's list

These papers were added to close gaps the team's list did not cover: evaluation methodology, calibration, cost-sensitive decisions, the origin of each dataset, and recent work (2025-2026) on leakage, hybrids and XAI.

Evaluation, leakage and metrics

Paper	Key finding	How we applied it	Where in FraudLens
Credit Card Fraud Detection: A Realistic Modeling and a Novel Learning Strategy. Pozzolo et al. (2018), IEEE Transactions on Neural Networks and Learning Systems, vol. 29, no. 8, pp. 3784-3797	Formalises the real-world FDS setting: concept drift, class imbalance and verification latency (only a small alerted subset is checked by investigators; most labels arrive days later). Proposes separating investigator feedback from delayed labels and evaluating with alert-based measures rather than plain accuracy.	Justifies the time-based train/val/test split and the use of precision-at-top-k / PR-AUC style metrics; frames delayed labels as future work for the online feature store.	Chronological split with label-delay gaps (config.py); PR-AUC & alert precision (metrics.py)
Learned lessons in credit card fraud detection from a practitioner perspective. Pozzolo et al. (2014), Expert Systems with Applications, vol. 41, no. 10, pp. 4915-4928	Practitioner-oriented comparison of imbalance handling, non-stationarity handling and evaluation metrics on real card data from a payment processor; highlights that metric choice and drift handling matter as much as the classifier.	Background for the imbalance-strategy comparison (none vs class-weight vs undersampling vs SMOTE) and for reporting several metrics.	E2 imbalance comparison; multi-metric evaluation
Reproducible Machine Learning for Credit Card Fraud Detection - Practical Handbook. Borgne et al. (2022), Universite Libre de Bruxelles (online book	Open, reproducible benchmark of essential fraud-detection techniques (simulated data, feature transformation, imbalance handling, ensembles, sequential/temporal validation, performance metrics) based on the ULB-Worldline collaboration; explicitly motivated by the lack of reproducibility in the field.	Template for the temporal validation protocol, delayed-label simulation and the choice of metrics; cite for the reproducibility argument.	7-day delay gaps (config.py); replay simulation (API stream)
The Precision-Recall Plot Is More Informative than the ROC Plot When Evaluating Binary Classifiers on Imbalanced Datasets. Saito & Rehmsmeier (2015), PLoS ONE, vol. 10, no. 3, e0118432	On strongly imbalanced data ROC curves can look deceptively good; precision-recall plots expose differences in false-positive behaviour that matter for the minority class.	Justifies reporting PR-AUC (average precision) as the primary metric and ROC-AUC only as secondary.	PR-AUC as the primary metric (metrics.py)
The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. Chicco & Jurman (2020), BMC Genomics, vol. 21, article 6	Accuracy and F1 can be over-optimistic on imbalanced data; MCC is high only when all four confusion-matrix cells are good.	Justifies adding MCC to the evaluation table and explains why '99.9% accuracy' claims in the literature are not informative.	MCC in every results table
Cross-Validation for Imbalanced Datasets: Avoiding Overoptimistic and Overfitting Approaches. Santos et al. (2018), IEEE Computational Intelligence Magazine, vol. 13, no. 4, pp. 59-76	A frequent experimental flaw is applying oversampling to the entire dataset before cross-validation, producing biased models and overly optimistic estimates; oversampling must be done inside each training fold only.	Primary methodological citation for the 'SMOTE-before-split inflates results' demonstration.	E3 protocols; resampling only inside the training pipeline

Paper	Key finding	How we applied it	Where in FraudLens
Leakage in data mining: Formulation, detection, and avoidance. Kaufman et al. (2012), ACM Transactions on Knowledge Discovery from Data, vol. 6, no. 4, article 15, pp. 1-21	Formal definition of leakage (information about the target illegitimately available at training time), with detection and avoidance methodology; recommends 'learn-predict separation' respecting time.	Supports computing behavioural features only from past transactions (online feature store) and the time-ordered split.	Causal feature store + online/offline parity test
Fraud Dataset Benchmark and Applications. Grover et al. (2023), arXiv preprint arXiv:2208.14417 (v1 Aug 2022	FDB compiles public fraud datasets (including card-not-present fraud) with standardised train/test splits and demonstrates feature engineering, supervised-model comparison, label-noise removal, class-imbalance treatment and semi-supervised learning.	Reference for standardised benchmarking and for the observation that fraud data have unique properties (imbalance, drift, adversarial).	Benchmarking discipline; candidate future datasets
Data Leakage and Deceptive Performance: A Critical Examination of Credit Card Fraud Detection Methodologies. Hayat & Magnier (2025), Mathematics (MDPI), vol. 13, no. 16, article 2563	Identifies four flaws in card-fraud papers: leakage from preprocessing/oversampling before the split, vague reporting, no temporal validation, and recall optimisation at precision's expense; a minimal MLP with leakage reaches 99.9% recall, while a corrected pipeline gives modest but reliable metrics.	Direct peer-reviewed support for the SMOTE-before-split + random-split inflation experiment and the temporal split.	E3 reproduces the leakage effect they describe

Imbalance, calibration and cost-sensitive decisions

Paper	Key finding	How we applied it	Where in FraudLens
Calibrating Probability with Undersampling for Unbalanced Classification. Pozzolo et al. (2015), 2015 IEEE Symposium Series on Computational Intelligence (SSCI), pp. 159-166	Undersampling the majority class shifts the posterior probabilities of a classifier; the paper derives a correction (Bayes-rule adjustment) so that probabilities learned on undersampled data are re-calibrated to the true class prior. This is the paper the Kaggle ULB dataset page cites as the origin of the 284,807-transaction / 492-fraud European dataset.	Cite as the dataset source. Motivates why probabilities from models trained with undersampling/SMOTE must be re-calibrated before a Low/Medium/High risk score is thresholded.	ULB dataset source (E6); calibrated stacking output (hybrid.py)
SMOTE: Synthetic Minority Over-sampling Technique. Chawla et al. (2002), Journal of Artificial Intelligence Research, vol. 16, pp. 321-357	Synthetic minority oversampling combined with majority undersampling improves classifier sensitivity to the minority class on imbalanced data.	Original reference for the SMOTE arm of the imbalance comparison.	SMOTE arm of E2 and protocols A-C of E3
The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression. Goorbergh et al. (2022), Journal of the American Medical Informatics Association, vol. 29, no. 9, pp. 1525-1534	Random undersampling, oversampling and SMOTE produce strongly miscalibrated (over-estimated) risk probabilities without improving discrimination; imbalance correction is not needed when the output is a calibrated risk score with a chosen threshold.	Key argument for the calibrated risk score + cost-based threshold design over resampling, and for the calibration step after class weighting.	E2 calibration (Brier) results; stacking instead of resampling
The Foundations of Cost-Sensitive Learning. Elkan (2001), Proceedings of the 17th International Joint Conference on Artificial Intelligence (IJCAI-01), pp. 973-978	Optimal decisions under a cost matrix reduce to thresholding a calibrated probability at p* = C_FP/(C_FP+C_FN); changing class balance in training mainly shifts the threshold and can be corrected analytically.	Theoretical basis for the cost-based Approve / Step-up / Block thresholds on the calibrated fraud probability.	Cost-optimal alert threshold (metrics.cost_optimal_threshold)

Paper	Key finding	How we applied it	Where in FraudLens
Predicting good probabilities with supervised learning. Niculescu-Mizil & Caruana (2005), Proceedings of the 22nd International Conference on Machine Learning (ICML '05), pp. 625-632	Boosted trees push probabilities away from 0 and 1 (sigmoid distortion) while bagged trees/random forests are better calibrated; Platt scaling and isotonic regression fix the distortion.	Justifies calibrating the XGBoost/stack output (Platt or isotonic) before mapping to Low/Medium/High.	Calibration figure; logistic recalibration of boosted-tree scores
Predict-then-optimize or predict-and-optimize? An empirical evaluation of cost-sensitive learning strategies. Vanderschueren et al. (2022), Information Sciences, vol. 594, pp. 400-415	Compares cost-sensitive training (predict-and-optimize) with cost-sensitive thresholding of a standard model (predict-then-optimize) across fraud and credit datasets; cost-sensitive decision rules on a well-trained model are a strong, simple option.	Supports the 'train normally, then set cost-based thresholds' approach adopted in the risk engine.	Train normally, then set cost-based thresholds (hybrid.fit_meta)
Instance-dependent cost-sensitive learning for detecting transfer fraud. Hoppner et al. (2022), European Journal of Operational Research, vol. 297, no. 1, pp. 291-300	Fraud detection is an instance-dependent cost-sensitive problem (the cost of a missed fraud equals the transaction amount); proposes cost-sensitive logistic regression / boosting optimising expected savings rather than AUC.	Justifies weighting the cost of false negatives by transaction amount when choosing thresholds and reporting 'savings' alongside F1.	Missed-fraud cost = transaction amount (metrics.money_metrics)

Behavioural features and sequences

Paper	Key finding	How we applied it	Where in FraudLens
Feature engineering strategies for credit card fraud detection. Bahnsen et al. (2016), Expert Systems with Applications, vol. 51, pp. 134-142	Transaction-aggregation features (per-card counts/amounts over time windows) and periodic time-of-day features modelled with the von Mises distribution improve detection; evaluation uses a cost-based (financial) measure rather than misclassification rate; code released.	Directly supports the per-card velocity counts, amount-vs-history ratios and night-time flags, and the cost-based thresholding.	Velocity and amount-vs-history features (features.py); money-based thresholds
Transaction aggregation as a strategy for credit card fraud detection. Whitrow et al. (2009), Data Mining and Knowledge Discovery, vol. 18, no. 1, pp. 30-55	Introduces transaction aggregation (summarising an account's recent activity into features) as a fraud-detection strategy and studies the effect of aggregation window length across classifiers on real bank data.	Foundational citation for the behavioural aggregation features and for choosing window lengths (time since last transaction, rolling counts).	1 h / 24 h / 7 d aggregation windows (features.py)
Sequence classification for credit-card fraud detection. Jurgovsky et al. (2018), Expert Systems with Applications, vol. 100, pp. 234-245	Frames fraud detection as sequence classification with LSTMs over per-card transaction histories; LSTM improves detection on face-to-face transactions versus a random forest, while feature aggregation strategies remain highly effective for both.	Supports using per-card history features with tree models now, and LSTM sequence models as future work.	Velocity windows capture sequence signal; LSTM in future scope

Models and ensembles

Paper	Key finding	How we applied it	Where in FraudLens
XGBoost: A Scalable Tree Boosting System. Chen & Guestrin (2016), Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD '16), pp. 785-794	Scalable regularised gradient-boosted trees with sparsity-aware split finding and weighted quantile sketch.	Reference for the XGBoost base learner (and scale_pos_weight class weighting).	Main classifier (hybrid.py)
LightGBM: A Highly Efficient Gradient Boosting Decision Tree. Ke et al. (2017), Advances in Neural Information Processing Systems 30 (NIPS 2017)	Gradient-based One-Side Sampling and Exclusive Feature Bundling speed up GBDT training by up to 20x with near-identical accuracy.	Alternative gradient-boosting base learner; cite if LightGBM is benchmarked or used for latency reasons.	LightGBM benchmark (E1)
Random Forests. Breiman (2001), Machine Learning, vol. 45, no. 1, pp. 5-32	Bagged decision trees with random feature selection at each split; generalisation error converges as trees are added and depends on tree strength and inter-tree correlation.	Reference for the Random Forest base learner.	Second-opinion classifier (hybrid.py)
Stacked generalization. Wolpert (1992), Neural Networks, vol. 5, no. 2, pp. 241-259	Introduces stacking: a higher-level generaliser learns to combine the outputs of lower-level models, using held-out predictions to avoid bias.	Reference for the stacking meta-learner combining XGBoost, RF and Isolation Forest scores (train the meta-learner on out-of-fold / later-time predictions).	Logistic meta-learner fitted on the later validation period (hybrid.fit_meta)
Why do tree-based models still outperform deep learning on typical tabular data?. Grinsztajn et al. (2022), Advances in Neural Information Processing Systems 35 (NeurIPS 2022), pp. 507-520 (Curran)	Benchmark of 45 tabular datasets: tree-based models (XGBoost, random forests) remain state of the art on medium-sized tabular data, even before accounting for their speed.	Justifies choosing tree ensembles over the CNN/LSTM approaches in team papers 1, 2, 9 and 10.	E1: tree ensembles vs MLP / 1D-CNN
Tabular data: Deep learning is not all you need. Shwartz-Ziv & Armon (2022), Information Fusion, vol. 81, pp. 84-90	XGBoost outperforms recent deep tabular models across datasets (including those the deep models were proposed on) and needs far less tuning; an ensemble of deep models and XGBoost does best.	Second, independent justification for the tree-based hybrid design and for ensembling.	E1: tree ensembles vs deep models; ensembling
An Intelligent Approach to Credit Card Fraud Detection Using an Optimized Light Gradient Boosting Machine. Taha & Malebary (2020), IEEE Access, vol. 8, pp. 25579-25587	Bayesian-optimised LightGBM for card fraud detection evaluated on two public datasets (including the ULB dataset); reports strong accuracy/AUC versus other classifiers.	Comparator for gradient-boosting results on the ULB dataset; supports hyper-parameter optimisation of the boosting model.	Comparator for E6 (ULB)
Credit Card Fraud Detection Using State-of-the-Art Machine Learning and Deep Learning Algorithms. Alarfaj et al. (2022), IEEE Access, vol. 10, pp. 39700-39715	Comparative analysis of ML (ELM, DT, RF, SVM, LR, XGBoost) and CNN-based DL architectures on the European (ULB) benchmark dataset.	Widely cited ML-vs-DL comparator on the same ULB dataset.	ML-vs-DL comparator on ULB (E6)

Hybrid and anomaly detection

Paper	Key finding	How we applied it	Where in FraudLens
Combining unsupervised and supervised learning in credit card fraud detection. Carcillo et al. (2021), Information Sciences, vol. 557, pp. 317-331 (online 2019)	A hybrid that feeds unsupervised outlier scores, computed at several granularity levels (global, per card, per cluster), into a supervised classifier improves detection accuracy on a real annotated card dataset.	The closest published precedent for feeding Isolation Forest anomaly scores into the stacking meta-learner alongside XGBoost/RF outputs.	Isolation-Forest score in the stacking layer (hybrid.py); E4 'IF score as XGB feature'
Isolation Forest. Liu et al. (2008), 2008 Eighth IEEE International Conference on Data Mining (ICDM), pp. 413-422	Anomalies are 'few and different' and are isolated with fewer random partitions; linear-time, low-memory unsupervised anomaly detector.	Reference for the unsupervised anomaly branch of the stack.	Anomaly branch trained on genuine transactions only (hybrid.py)
A Neural Network Ensemble With Feature Engineering for Improved Credit Card Fraud Detection. Esenogho et al. (2022), IEEE Access, vol. 10, pp. 16400-16407	LSTM base learners in an AdaBoost ensemble with SMOTE-ENN resampling and feature engineering for imbalanced card fraud data.	Comparator for ensemble + feature-engineering designs; same first-author group as team paper 19.	Comparator: ensemble + feature engineering
A Hybrid Anomaly Detection Framework Combining Supervised and Unsupervised Learning for Credit Card Fraud Detection. Shanaa & Abdallah (2025), F1000Research, vol. 14, article 664 (v2, Dec 2025)	Autoencoder reconstruction error (trained on normal transactions) fused with an XGBoost score via optimised thresholding on the ULB dataset (284,807 transactions, 492 frauds): recall 0.925, precision 0.957, F1 0.941, MCC 0.941; code and pipeline released.	Closest recent comparator to the XGBoost + anomaly-score hybrid on the same dataset; use its reported numbers as a benchmark.	Closest published comparator for the ULB hybrid (E6)

Explainable AI

Paper	Key finding	How we applied it	Where in FraudLens
A Unified Approach to Interpreting Model Predictions. Lundberg & Lee (2017), Advances in Neural Information Processing Systems 30 (NIPS 2017)	SHAP unifies six additive feature-attribution methods and proves a unique solution with desirable properties (local accuracy, missingness, consistency).	Reference for SHAP values behind the reason codes.	SHAP reason codes (explain.py)
From local explanations to global understanding with explainable AI for trees. Lundberg et al. (2020), Nature Machine Intelligence, vol. 2, no. 1, pp. 56-67	TreeSHAP: polynomial-time exact Shapley values for tree ensembles, interaction values, and tools to build global understanding from many local explanations.	Reference for computing exact, fast SHAP values for XGBoost/RF at API time and for global feature-importance plots on the dashboard.	Exact TreeSHAP in the API + global importance figure
Explainable Machine Learning for Fraud Detection. Psychoula et al. (2021), Computer (IEEE), vol. 54, no. 10, pp. 49-59	Explores explainability methods for real-time fraud detection, examining the choice of SHAP background datasets and run-time trade-offs on supervised and unsupervised models.	Supports the design decision to compute SHAP explanations inside the real-time API and to pick a background dataset carefully.	SHAP computed inside the real-time API (XGBoost pred_contribs)
Interpretable Ensemble Learning Models for Credit Card Fraud Detection. Iqbal et al. (2025), Applied Sciences (MDPI), vol. 15, no. 22, article 12073	Combines Random Forest and XGBoost with SHAP and LIME on a real-world card-fraud dataset, arguing that systematic XAI is scarce in fraud work and is required for regulatory compliance and trust.	Supports SHAP-based reason codes for RF/XGBoost outputs.	RF + XGBoost with SHAP (hybrid.py, explain.py)

Paper	Key finding	How we applied it	Where in FraudLens
AI-based credit card fraud detection: a machine learning approach with model explainability on real-world data. Cavdar & Bozanta (2026), Knowledge and Information Systems (Springer), vol. 68, no. 1	Nine supervised/unsupervised models under four sampling schemes (imbalanced, undersampling, SMOTE, voting+undersampling) on a private real dataset and a public one; Random Forest best among ML models, TabTransformer among DL; tuned XGBoost had a statistically higher F1 but RF was more robust without tuning; explained with feature importance, SHAP, LIME and DeepLIFT.	Supports keeping both RF and XGBoost in the stack and comparing sampling strategies; recent XAI precedent on real bank data.	Keep both RF and XGBoost; sampling comparison (E2)
Explainable ensemble machine learning for financial transaction fraud detection: Insights from XGBoost and deep neural networks. Luong & Xie (2026), The Journal of Finance and Data Science, vol. 12, article 100195	XGBoost + DNN ensemble using transactional, temporal and identity-linked features; evaluates focal loss, class weighting and threshold optimisation for imbalance; SHAP for interpretability on a real transaction dataset.	Supports class-weighting + threshold optimisation over resampling, and SHAP in a regulated setting.	Class weighting + threshold optimisation over resampling (E2)

Streaming, concept drift and feedback

Paper	Key finding	How we applied it	Where in FraudLens
Credit card fraud detection and concept-drift adaptation with delayed supervised information. Pozzolo et al. (2015), 2015 International Joint Conference on Neural Networks (IJCNN), pp. 1-8	Shows that investigator feedback and delayed labels must be handled separately; the winning strategy trains two classifiers (on feedbacks and on delayed labels) and aggregates them, substantially improving alert precision on real transactions.	Supports the two-model / aggregated-score design and explains how the system could be retrained under delayed labels in production.	Analyst feedback stored separately from dataset labels (api/db.py)
SCARFF: a scalable framework for streaming credit card fraud detection with Spark. Carcillo et al. (2018), Information Fusion, vol. 41, pp. 182-194	Integrates Kafka, Spark and Cassandra with an ML approach handling imbalance, non-stationarity and feedback latency; shown scalable, efficient and accurate on a massive real transaction stream.	Architecture reference for the FastAPI real-time API + online feature store; shows what a production streaming version would add.	Online feature store + real-time API; Kafka/Spark in future scope
Streaming active learning strategies for real-life credit card fraud detection: assessment and visualization. Carcillo et al. (2018), International Journal of Data Science and Analytics, vol. 5, no. 4, pp. 285-300	Journal extension of the DSAA 2017 paper (team paper 11): compares supervised, semi-supervised and unsupervised querying criteria for selecting which cardholders investigators should contact; High-Risk Querying is a strong baseline.	Supports the 'Step-up verification' tier as a human-in-the-loop querying mechanism whose labels can feed retraining.	STEP-UP tier and case queue (api/, dashboard /cases)
Incremental learning strategies for credit cards fraud detection. Lebichot et al. (2021), International Journal of Data Science and Analytics, vol. 12, no. 2, pp. 165-174	Compares incremental (sliding-window / ensemble) learning strategies for adapting fraud detectors to concept drift on real Worldline data.	Future-work citation for periodic retraining of the stack as fraud patterns drift.	Future scope: incremental retraining from reviewed cases
A survey on concept drift adaptation. Gama et al. (2014), ACM Computing Surveys, vol. 46, no. 4, article 44, pp. 1-37	Taxonomy of concept-drift types and adaptive-learning strategies (detection, sliding windows, ensembles) and of evaluation methodology for evolving data.	Background for the drift discussion and for evaluating on a later time slice.	Monthly stability and PSI drift check (E7)

Paper	Key finding	How we applied it	Where in FraudLens
A drift adaptive framework for detecting and tracking evolving anomalies in financial transaction streams. Thiong'o et al. (2026), Scientific Reports (Nature)	HADA+ hybrid anomaly-detection architecture with drift-responsive prioritisation and lifecycle tracking; on the (ULB) Credit Card Fraud dataset reports ROC-AUC 0.915 and PR-AUC 0.069 for the unsupervised setting, illustrating how much lower PR-AUC is than ROC-AUC on this data.	Recent evidence for concept-drift handling and for the ROC-vs-PR gap on the ULB dataset; supports using PR-AUC.	ROC-vs-PR gap on ULB; drift monitoring (E7)

Graph learning (future work)

Paper	Key finding	How we applied it	Where in FraudLens
Pick and Choose: A GNN-based Imbalanced Learning Approach for Fraud Detection. Liu et al. (2021), Proceedings of the Web Conference 2021 (WWW '21), pp. 3168-3177	PC-GNN: label-balanced sampling of nodes/edges and neighbour choosing to counter class imbalance in graph-based fraud detection.	Future-work citation for graph-based extension using card-merchant-device relations.	Future scope: card-merchant graph model

6 Our results next to published claims

Table 13. Published numbers versus ours (reported numbers are taken from the papers' abstracts)

Paper	Dataset	Their reported result	Our comparable result	Comment
T3 Jayasingh & Sri 2023	ULB	XGBoost accuracy 99.96%, recall 83%	XGBoost recall 0.760, precision 0.905, accuracy 99.96%; hybrid recall 0.760	Same dataset. We split by time and tune the threshold on validation, so our numbers are harder to earn.
T7 Gupta et al. 2024	not named	precision 99.94%, accuracy 99.97%	E3 protocol A reproduces 99.96% precision on ULB only when SMOTE is applied before the split	Such precision is only reached with synthetic fraud in the test set; honest protocol D gives 88.7%.
T1 Banoth & Madhavi 2024	Kaggle (unnamed)	CNN accuracy 99.87%	Our 1D-CNN accuracy 99.84%, PR-AUC 0.933 (Sparkov)	Accuracy is uninformative: 'always genuine' already scores 99.40%.
T10 Zhang & Huang 2020	ULB	~93% accuracy (CNN/AE + balancing)	Our MLP on ULB: PR-AUC 0.809, F1 0.812	Tree ensembles outperform the DL baselines on the same split (E1, E6).
Shanaa & Abdallah 2025	ULB	AE + XGBoost hybrid: recall 0.925, precision 0.957, F1 0.941	Hybrid: recall 0.760, precision 0.905, F1 0.826	Closest design. Differences in split protocol (theirs is not chronological) make direct comparison indicative only.
Thiong'o et al. 2026	ULB	Unsupervised PR-AUC 0.069 (ROC-AUC 0.915)	Our Isolation Forest (unsupervised) PR-AUC 0.051	Confirms that unsupervised detectors alone are weak on ULB; they belong inside a hybrid.
Hayat & Magnier 2025	ULB	Leaky pipeline: 99.9% recall; corrected: modest	E3: F1 1.000 (leaky) vs 0.803 (honest)	Independent replication of the same leakage effect.

How we prove it

Numbers from different papers are rarely comparable: datasets, splits, thresholds and metrics differ. That is why FraudLens re-implements the competing ideas (CNN, MLP, SMOTE + XGBoost, simple ML, rules) and tests them on one shared chronological split with the same metrics (E1-E3).

7 Industry and regulatory context

Table 14. Context used in the introduction of the Project Report

Topic	Figure	Source	Date
Global payment-card fraud losses (issuers, merchants, acquirers, ATM processors) in 2024	USD 33.41 billion in 2024 (down 1.2% year on year) on USD 51.920 trillion of card volume; 6.43 cents per USD 100 (down from 6.58 cents in 2023); the US accounted for 26.31% of card volume but 41.87% of fraud losses; projected USD 41.06 billion by 2030 and USD 48.50 billion by 2034 (cumulative USD 407.60 billion over the next 10 years)	The Nilson Report, press release 'Global Card Fraud Losses at $33 Billion' (GlobeNewswire) (https://www.globenewswire.com/news-release/2026/01/07/3214821/0/en/global-card-fraud-losses-at-33-billion.html)	2026-01-07
Long-run growth of global card fraud losses (historical trend)	USD 9.84 billion (2011) to USD 27.85 billion (2018), projected to exceed USD 40 billion by 2027 (2019 Nilson Report as quoted in the ULB handbook foreword); the same foreword notes the fraud-loss percentage began to decline in 2016 alongside adoption of ML-based detection	Le Borgne et al., Reproducible Machine Learning for Credit Card Fraud Detection - Practical Handbook, Foreword (citing Nilson Report 2019) (https://fraud-detection-handbook.github.io/fraud-detection-handbook/Foreword.html)	2022
Payment fraud in the European Economic Area, 2024	Total reported payment fraud EUR 4.2 billion in 2024 (EUR 3.5 billion in 2023, EUR 3.4 billion in 2022); card-payment fraud losses on EU/EEA-issued cards EUR 1.329 billion in 2024 (+29% y/y); overall fraud rate about 0.002% of transaction value; card fraud was 17 times higher when the payee was outside the EEA where strong customer authentication is not required	EBA-ECB joint 2025 Report on Payment Fraud, ECB press release (https://www.ecb.europa.eu/press/pr/date/2025/html/ecb.pr251215~e133d9d683.en.html)	2025-12-15
Card-not-present (CNP) fraud share of total card fraud value (Europe)	In 2019, 80% of the value of card fraud on SEPA-issued cards came from CNP transactions (internet, mail, phone) - EUR 1.50 billion of EUR 1.87 billion total; point-of-sale fraud 15% and ATM fraud 5%	European Central Bank, Seventh report on card fraud (https://www.ecb.europa.eu/press/cardfraud/html/ecb.cardfraudreport202110~cac4c418e8.en.html)	2021-10
UK unauthorised fraud and remote-purchase (CNP) fraud, 2024	3.13 million unauthorised fraud cases (+14%) and GBP 722 million losses (+2%) in 2024; remote purchase fraud cases +22% to nearly 2.6 million and losses +11% to just under GBP 400 million, driven partly by social engineering of one-time passcodes; card ID theft losses fell 26% to GBP 58.7 million; GBP 1.45 billion of unauthorised fraud was prevented (67p per GBP 1 attempted); customers refunded in >98% of unauthorised cases	UK Finance, Annual Fraud Report 2025 press release (https://www.ukfinance.org.uk/news-and-insight/press-release/fraud-report-2025-press-release)	2025-05-28

Topic	Figure	Source	Date
India: bank frauds reported to RBI (frauds of Rs 1 lakh and above), latest three years	2025-26: 10,114 cases / Rs 48,021 crore; 2024-25: 23,722 cases / Rs 32,803 crore; 2023-24: 35,800 cases / Rs 11,013 crore (Table VI.2). Card/Internet/Digital-payments category: 28,836 cases (80.4% of all cases) / Rs 1,452 crore in 2023-24; 13,332 cases (56.2%) / Rs 517 crore in 2024-25; 293 cases (2.9%) / Rs 29 crore in 2025-26 (Table VI.3). RBI notes card/internet frauds were the most numerous category in 2023-24 and 2024-25, and that 2025-26 data include 314 re-classified older cases worth Rs 30,199 crore	Reserve Bank of India, Annual Report 2025-26, Chapter VI (Regulation, Supervision and Financial Stability), Tables VI.2 and VI.3 (https://www.rbi.org.in/Scripts/AnnualReportPublications.aspx?Id=1466)	2026-05-29
India: bank fraud figures as first published in the previous RBI Annual Report (before revisions)	2024-25: 23,953 cases / Rs 36,014 crore (amount roughly three times 2023-24's Rs 12,230 crore; case count down from 36,060) - secondary reporting of RBI Annual Report 2024-25; the 2025-26 report later shows revised 2024-25 figures of 23,722 cases / Rs 32,803 crore	RBI Annual Report 2024-25 as reported by Deccan Herald and SBI Ecowrap (secondary sources; the RBI PDF is behind a bot-check) (https://www.deccanherald.com/business/bank-frauds-value-trebles-to-over-rs-36000-crore-3563349)	2025-05-29
India: customer liability rules for unauthorised electronic banking transactions	RBI circular RBI/2017-18/15, DBR.No.Leg.BC.78/09.07.005/2017-18 'Customer Protection - Limiting Liability of Customers in Unauthorised Electronic Banking Transactions': zero customer liability for bank negligence or third-party breach reported within 3 working days; limited liability (max Rs 10,000 for credit cards with limit up to Rs 5 lakh, Rs 25,000 above) if reported within 4-7 working days; banks must shadow-credit the amount within 10 working days and resolve within 90 days; burden of proof lies on the bank; banks must have a 'robust and dynamic fraud detection and prevention mechanism' and mandatory SMS alerts; distinguishes CNP/remote from card-present transactions	Reserve Bank of India notification (https://www.rbi.org.in/commonman/english/scripts/Notification.aspx?Id=2336)	2017-07-06
India: regulatory requirements for digital payment security and fraud risk management	RBI Master Direction on Digital Payment Security Controls (RBI/2020-21/74, DoS.CO.CSITE.SEC.No.1852/31.01.015/2020-21), effective six months after issue: para 36-37 require documented rules and parameterised alerts on transaction velocity, high-risk merchant category codes, counterfeit-card/CVV patterns, new-account activity, time zones, geo-locations, IP origin and behavioural biometrics; para 34 allows adaptive (risk-based) authentication; para 41 requires near-real-time reconciliation; para 72 requires 24x7 monitoring of card transactions with card/BIN-level limits; para 39 requires staff able to 'differentiate false positives from the rest'	Reserve Bank of India, Master Direction on Digital Payment Security Controls (https://www.rbi.org.in/Scripts/BS_ViewMasDirections.aspx?id=12032)	2021-02-18
ULB/Worldline European credit-card dataset (Kaggle mlg-ulb/creditcardfraud) size and imbalance	284,807 transactions with 492 frauds (about 0.172%); PCA-anonymised features V1-V28 plus Time and Amount; originates from the ULB Machine Learning Group - Worldline collaboration (Dal Pozzolo et al., 2015)	Figures confirmed in peer-reviewed abstracts using the dataset (Shanaa & Abdallah 2025, F1000Research 14:664; Jayasingh & Sri 2023, ESCI); Kaggle page itself could not be fetched automatically (https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud)	2025

Topic	Figure	Source	Date
Sparkov simulated dataset (Kaggle kartik2112/fraud-detection) provenance	Generated with the open-source 'Sparkov Data Generation' synthetic credit-card transaction generator (Brandon Harris, namebrandon/Sparkov_Data_Generation, originally based on code by Josh Plotkin); v0.3 of the generator introduced fraudulent transactions/profiles and Gamma-distributed amounts; customer profiles, merchant names and spending-category windows are simulated	GitHub repository README (Kaggle page could not be fetched automatically; row counts of fraudTrain/fraudTest not verified here) (https://github.com/namebrandon/Sparkov_Data_Generation)	accessed 2026-09-11

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